MASTER GAME DESIGN DOCUMENT

FUSE VALLEY

Tag, but you're holding a bomb.
Pass it on. The fuse never resets.



DOCUMENT
Master Game Design Document

VERSION
1.0 · 2026-10-17

PLATFORM
Roblox · PC · Mobile · Console

BUILT WITH
Rojo · Luau (strict) · Claude Code

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Version 1.0 · 2026-10-17 · Name: FUSE VALLEY (rename = change `Config.Game.NAME` + store page)

> A 20-player Roblox tag game with a ticking bomb. Runners sprint across a field to the opposite safe side while Bombers hunt them. A Bomber's touch sticks a live bomb to your back. You can pass it on with a bump, but **the fuse never resets**. If it blows on you, you join the Bombers. The last runner standing wins.
>
> The crossing formula is the playground-tag loop players already know from *Huss Valley*. The bomb, the fuses, the presentation, the progression and every asset are ours.

0. How to use this document (read first — humans and Claude Code)

1. **This document is the single source of truth.** If something is not written here, do not build it. If two sections disagree, the more specific one wins, and the conflict is logged in `docs/DECISIONS.md`.
2. **Every feature has an ID**, for example `F-BOMB-04`. Code header comments, commits and tests reference these IDs.
3. **No magic numbers.** Every number lives in a `Config` module (Appendix A). Code reads the config and never hard-codes values.
4. **Tags tell you where a rule comes from:**
   - PARITY: the proven crossing-tag rules, kept one-for-one.
   - OURS: our addition or decision.
   - TUNE: a starting value expected to change after playtests.
5. **Build in phase order (§26).** Each phase has a Definition of Done. Never start phase N+1 until phase N's DoD passes.
6. **Assets** (models, sounds, UI art, VFX) are specified in the companion **ASSET BIBLE**. Code references assets only through `Shared/Assets.luau` keys, never raw IDs.
7. **The premium bar is not optional.** Read Asset Bible §1 ("The Quality Bar") before making *anything* players can see or hear. If an asset fails the checklist there, it does not ship, not even as a placeholder.
8. **Simplicity test.** Every feature must pass both checks:
   - (a) It uses the existing controls.
   - (b) It can be explained in one line of on-screen text.

   If a feature fails either check, it is cut or redesigned.

0.1 Document map

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1. Vision

1.1 The one-liner

Tag, but you're holding a bomb. Get tagged and a bomb sticks to your back, beeping faster and faster. Bump someone to pass it, but the clock never resets.

1.2 Pillars (every decision is checked against these)

#	Pillar	What it means in practice
P1	**Panic you can see**	The danger is always physical and visible: a real bomb on a real back, a fuse that visibly burns down, a field that glows before it blows. Nobody ever has to read a number to feel the danger.
P2	**Three-second stories**	Every pass, defuse and explosion is a tiny story with a twist (betrayal, revenge, clutch). We design for those moments and celebrate every one of them.
P3	**Never wait, never empty-handed**	Blown up? You're a Bomber instantly and still playing. Every round pays something. There's a reward beat every ~10 seconds, a level-up every ~10 minutes, and a gift every ~15 minutes.
P4	**Premium or nothing**	Every model, sound, effect and screen is crafted to the Quality Bar (Asset Bible §1). No toolbox models, no AI-generated art, no generic gradients, no emoji icons, no stock "game UI kit" look.
P5	**Simple to learn, deep to master**	Four rules for the bomb. Mastery comes from movement (dash, slide, hard stop), pass timing, reading the fuse and positioning, never from extra buttons.
P6	**Mobile-first**	Most players are on phones. Every rule works with a thumbstick and auto-pass on contact. Every HUD element is laid out for a 844×390 screen first.

1.3 Non-goals (do not build)

- No pay-to-win of any kind. Nothing purchasable changes speed, fuse time, hitbox or anything else that affects the outcome of a round.
- No weapons, health bars, abilities or items in the standard round. The bomb *is* the game.
- No guns, gore or realistic violence. Explosions are cartoon pyrotechnics: puffs, sparks, confetti-adjacent debris and soot. Bodies fling and tumble; they never break.
- No trading at launch.
- No custom chat.

1.4 Target experience

Moment	Player feeling	Delivered by
First 10 seconds	"I get it."	Bombs are visible on backs, the 4-rule tutorial card (§20) and the first crossing
First tag on you	Pure panic, then a choice	Heartbeat, red vignette, fuse spark, the pass prompt (§10)
A pass with 0.5 s left	Laughter or rage, both shareable	Slow-mo, callout, kill-cam replay (§10.5)
Last Stand	Everyone watching one person	Spotlight, drumroll, crowd chant (§8.2)
Results	"One more."	XP bar, quest ticks, "1 more round to level up" (§8.4)

2. Locked decisions (2026-10-17)

These were the open questions before writing this GDD. The user said "go" with the recommended picks; each is changeable through the noted config key.

#	Decision	Value	Config
D1	Name	**FUSE VALLEY** (was the working title FUSE VALLEY; the user named the place on 2026-10-17)	`Config.Game.NAME`
D2	Max players per server	**30** (raised from 20 on 2026-10-18, the user's call)	`Config.Game.MAX_PLAYERS`
D3	Bots fill each round to	**16 participants**	`Config.Bots.FILL_TARGET`
D4	Starting Bombers	**1 per 8 participants** (min 1, max 3) → 2 in a 16-player round	`Config.Round.BOMBERS_PER`
D5	Bomb fuse	**4.0 s**	`Config.Bomb.FUSE`
D6	Max crossings	**8**	`Config.Round.MAX_CROSSINGS`
D7	Engine plumbing (networking, saving, participants, game state)	**Transferred** from Chase Game	§23
D8	Bots	**Yes**: Chase Game's bot movement, new bomb-aware brains	§18
D9	Roblox owner	**Same group as Chase Game** (keeps published animations and map props working)	§23.4
D10	Team split	**Changed 2026-10-17:** Partner: all gameplay scripting · User: UI, maps, assets (art, VFX, audio, models) · Claude Code builds both from this document	`docs/ART_BACKLOG.md`
D11	Art direction	**New: "Arcade Pyrotechnic"** (Asset Bible §2). It does not use Chase Game's studded look.	Asset Bible

3. Glossary

Term	Meaning
Runner	A player trying to cross. Starts as the majority team.
Bomber	A hunter. Their touch sticks a bomb onto a Runner. Bombers win if every Runner is blown up.
Crossing	One run from one Safe Strip to the other. A round has up to 8 crossings, alternating direction.
Safe Strip	The zone at each end of the field. Runners inside can't be tagged or bumped. A bomb carried in is defused.
Field	The play area between the two Safe Strips. Bombers can't leave it.
Bomb	The live explosive stuck to a player's back. It has a fuse with a remaining time in seconds.
Carrier	The player currently holding the bomb.
Pass	A Carrier bumps another player and the bomb jumps to them, keeping its remaining time.
Fuse	The bomb's remaining time. It never resets on a pass.
Field Fuse	The crossing's own timer. At 0 the whole field detonates.
Boom	A bomb exploding on its Carrier.
Defuse	Carrying a bomb into a Safe Strip. It fizzles harmlessly.
Clutch Defuse	A defuse with ≤ 1.0 s left on the fuse.
Revenge Pass	Passing a bomb onto a Bomber.
Betrayal	Passing a bomb to a Runner who then booms with it.
Chain	One bomb passing through 3+ Carriers before it booms or is defused.
Last Stand	The final solo crossing of the last Runner.
MEGA BOMB	The 15-minute scheduled event (§16).
Stars	The soft currency (§13).
Participant	A human or bot in the round. Every gameplay function takes a Participant.

4. Parity map — what we keep vs what is ours

Area	Rule	Source
Field	Two Safe Strips, a field between them, crossings alternate direction	PARITY
Hunters	Hunters start in the middle and can't enter the Safe Strips	PARITY
Elimination	A caught Runner joins the hunters and keeps playing	PARITY
Win	Last Runner standing wins	PARITY
Elimination method	Not instant. The tag sticks a 4 s bomb that can be passed or defused	OURS
Stalling	The Field Fuse detonates the field when the crossing timer ends	OURS
Hunters getting hit	A bomb passed onto a Bomber stuns them for 3 s	OURS
Finale	The Last Stand solo crossing with spotlight	OURS
Movement	Dash, slide, hard stop, omnimovement (from Chase Game)	OURS
Event	MEGA BOMB every 15 minutes	OURS
Progression, store, art, audio, UI	Original	OURS

5. Match flow & state machine

5.1 The loop at a glance

```
LOBBY/INTERMISSION (12 s) → INTRO (4 s) → CROSSING 1 → CROSSING 2 → … → CROSSING 8
                                                  |                (ends early: 0 Runners, or 1 Runner → LA
ST STAND)
                                                  ▼
                                        LAST STAND → RESULTS (8 s) → INTERMISSION
```

A full round takes **2–3 minutes** TUNE. There is no map vote at launch (one map). This keeps the time between rounds at 20 seconds total (F-FLOW-01).

5.2 Phases (server state machine)

`Types.Phase`:

ID	Phase	Duration	What happens	Exit
F-FLOW-02	`INTERMISSION`	12 s TUNE	Everyone in the lobby. The shop, quests and gifts are open, and the MEGA BOMB clock is visible. A 3-2-1 plays in the last 3 s.	Timer
F-FLOW-03	`INTRO`	4 s	Teleport to the start Safe Strip. A camera sweep shows the Bombers in the middle with role chips. The bombers are revealed with a "BOMBERS" stinger.	Timer
F-FLOW-04	`READY`	2 s	Runners stand behind the start line (a holding barrier), and the Field Fuse appears at full time. "3, 2, 1, GO" (shortened to "2, 1, GO" from crossing 2).	Timer
F-FLOW-05	`CROSSING`	Field Fuse (§6.2)	Barrier drops. Runners run; Bombers tag; bombs pass and boom.	All Runners safe, OR Field Fuse = 0, OR Runners ≤ 1
F-FLOW-06	`RESOLVE`	1.5 s	Field detonation (if the fuse hit 0), pending booms finish, and the turn-over stinger plays ("+2 BOMBERS").	Timer
F-FLOW-07	`LAST_STAND`	Intro 3 s + Field Fuse 12 s	§8.2	Runner safe or boomed
F-FLOW-08	`RESULTS`	8 s	Winner reveal, podiums, the XP bar, quest ticks (§8.4)	Timer

Who plays what (F-FLOW-09):

- **Bombers** spawn in the middle of the field at each `READY` and are held there until `GO`.
- **Runners** who reached the other side stay safe until the crossing ends.
- **Players joining mid-round** land in The Depot (§19.1) and drop in as **Bombers** at the next `READY` OURS (D-LOBBY-01). "YOU'RE UP NEXT" shows while they wait. Nobody waits a whole round, which is pillar P3.

5.3 Ending conditions (F-FLOW-10)

Checked continuously during `CROSSING`, in this order:

1. **0 Runners left** → Bombers win → `RESULTS`.
2. **1 Runner left**, and the round started with ≥ 4 Runners → finish the current crossing, then `LAST_STAND`.

   If the last Runner is already standing in a Safe Strip when they become the last one, the Last Stand starts after `RESOLVE`.
3. **The crossing completes** (every Runner safe or boomed, or the Field Fuse = 0):
   - If this was crossing 8 → the survivors win → `RESULTS`.
   - Otherwise, the next crossing starts in the other direction.

5.4 Round sizing

Participants	Starting Bombers	Field width	Notes
2–7	1	narrow (map variant)	With fewer than 4 Runners at the start, there is no Last Stand
8–15	1–2	standard	
16 (bot fill)	2	standard	The default case
17–23	2	standard	
24–30	3	standard	Bots stop filling at 16; above that every participant is a real player

Formula: `bombers = clamp(floor(n / BOMBERS_PER), 1, BOMBERS_MAX)`. The starting Bombers are chosen at random, but nobody starts as a Bomber two rounds in a row (`Config.Round.BOMBER_REPEAT_GUARD`).

6. Crossings & the Field Fuse

6.1 Crossing rules PARITY

- **F-CROSS-01:** Runners must leave the start Safe Strip when `GO` sounds.
  - A Runner still inside the start strip **5 s** after GO gets a red "MOVE!" warning (`Config.Round.IDLE_WARN`).
  - At the Field Fuse's end, **anyone not in the destination strip is on the field**, so they detonate (§6.2).
  - This covers anyone who stays behind in the start strip too. You cannot wait out a crossing.
- **F-CROSS-02:** Safe Strips block Bombers with a soft wall: they slide along it, with no hard bounce.
- **F-CROSS-03:** Inside a Safe Strip you **can't be tagged and can't receive a pass**. Bombs carried in are defused (§7.6).
- **F-CROSS-04:** A Runner who reaches the destination strip is **safe for the rest of the crossing**. They can't walk back onto the field (soft wall) until the next `READY`.

6.2 The Field Fuse (F-FUSE-01..05) OURS

The crossing's own timer is a **physical fuse**: a burning cord runs along both edges of the field, with a spark racing toward the far end. This is pillar P1: the timer is something you can see in the world, not just a number.

ID	Rule
F-FUSE-01	The Field Fuse lengths per crossing are `20, 16, 13, 11, 10, 10, 10, 10` s TUNE (`Config.Round.FIELD_FUSE`). The Last Stand uses `12` s.
F-FUSE-02	**Warning at 5 s:** the field floor's hazard lines glow, a klaxon plays and the HUD fuse turns red. **At 3 s:** the glow pulses and the music drops to a heartbeat.
F-FUSE-03	**At 0, the field detonates.** A rolling chain of blasts runs across the field over 0.6 s. Every participant still on the field booms: Runners join the Bombers, and Bombers are only flung (they respawn in the middle at `READY`).
F-FUSE-04	When every Runner is safe or boomed before the fuse ends, the crossing ends early and the remaining fuse fizzles out. You get a "ALL CLEAR" sting.
F-FUSE-05	Field detonation booms award **no** Bomber credit. The callout is "FUSE OUT" in grey; it's the field's kill, not a person's.
F-FUSE-06	**The hurry-up** (D-PACE-01): once half the Runners still in play are safe, or only idlers are left in the start strip, the Field Fuse cuts to its last **4 s** (`HURRY_SHARE`, `HURRY_LEFT`). "HURRY!" slams in and the 5-second warning fires at once. The Runners who made it never stand around waiting on stragglers, and the last ones get a scramble. The Last Stand keeps its full fuse.

6.3 Why the Field Fuse matters

- It removes the one bad pattern of crossing-tag games: Runners hugging the safe edge and waiting.
- It shortens every crossing as the round goes on, so the pace climbs exactly as the Bomber crowd grows.

6.4 Round modifiers (F-MOD-01..05) OURS

Some rounds bend one rule. A modifier has to pass all six of these tests, or it doesn't ship:

1. It makes tag, pass and defuse matter **more**, and never replaces them.
2. Every role gets new decisions, and nobody idles.
3. No death comes without a tell and a counter.
4. It can be explained in one line in 2 seconds at INTRO.
5. It keeps the Runner / Bomber balance, as measured by the pace probe.
6. It needs no new art.

ID	Rule
F-MOD-01	**Rotation:** every **3rd** round is a modifier round (`Config.Modifiers.EVERY`). The modifier is picked at random from the pool, never the same one twice in a row. **Never on a MEGA BOMB round:** MEGA wins, and the modifier waits for the next round. A modifier that needs more players (the Mole) is skipped when there aren't enough. INTRO announces it as the headline, with the modifier's name and its one-line rule. A small chip under the team counter names it for the whole round. Studio: `Config.Debug.FORCE_MODIFIER` .
F-MOD-02	**READ THE FACE**: *"No timers. Watch the bomb."* While a bomb is LIT, nothing shows its time: no overhead ring count, no burning braid, no tick speed-up, no faster Carrier vignette, no SAFE / TOO FAR verdict (the chevron shows only the run time). Every fuse is a random **3.0–5.0 s**, so the average stays 4.0. **The face never lies:** from **2.5 s left** (HOT) everything shows as usual. The face goes UHOH, and PANIC is always the last **1.0 s**. The server doesn't send the boom time to any client until the bomb turns HOT. Bots act on the same view: while a bomb is LIT, they assume the shortest fuse.
F-MOD-03	**LIGHTS OUT**: *"The Yard goes dark. Bombs are the only light."* The Yard goes to night, never pitch black: silhouettes still read on a phone. Every live bomb is a lamp in its stage colour, **16 / 20 / 24 studs** at LIT / HOT / CRITICAL. Every Runner has a soft **6-stud** glow. Bombers have no glow, only two red eye glints that fade in within **28 studs**. The Safe Strips glow. A Carrier lights up the Bombers around them, and themselves too. Bots see the same: bot Runners see a Bomber only within 28 studs or inside a bomb's light.
F-MOD-04	**THE MOLE**: *"One Runner is secretly a Bomber."* At INTRO, one Runner becomes the Mole, and only they are told. It needs **4+** starting Runners and is never a player in their first **5** rounds. The Mole plays and scores as a Runner. **STRIKE** (Q / gamepad X / the on-screen button) arms them for **3 s**: the next Runner they touch outside a Safe Strip gets a normal bomb, under the same rules as a tag. The Mole can't strike while carrying a bomb or standing in a Safe Strip, and a missed strike can be retried after **5 s**. A strike **reveals** the Mole: "IT WAS THE MOLE: name" goes to everyone, and the Mole is a Bomber from then on. The Mole is also revealed if they boom, at the end of crossing **3**, or once only one other Runner is left, so a Last Stand is never against a hidden Bomber. The strike counts as a tag, and its boom credits the Mole.
F-MOD-05	**Rewards and records:** modifiers change no payout table. Every Star still comes from the normal callouts and round payouts (F-ECO-03 cap). The pace probe reports every modifier separately (passes per bomb, boom %, who won).

Why these three: each one bends a different axis, so the rotation never feels like the same round again.

- READ THE FACE bends **timing**, and turns the hero bomb's face into the gameplay.
- LIGHTS OUT bends **what you can see**, and turns every bomb into a moving light.
- THE MOLE bends **who you trust**: Runners need each other to pass, and now they can't be sure of each other.

7. The Bomb (core mechanic)

7.1 The four rules (this is the whole on-screen tutorial)

1. **A Bomber's touch sticks a bomb on you.**
2. **Bump anyone to pass it.**
3. **The fuse never resets.**
4. **Reach a safe side to defuse it.**

7.2 The physical bomb (F-BOMB-01)

- A real 3D bomb model (Asset Bible §3) is **mounted on the Carrier's back** on a sprung clamp. It sits 0.4 studs off the torso, so it jiggles with movement.
- The fuse **is** the timer:
  - The braided fuse cord on top visibly burns down from full length to the cap.
  - The fuse length shown always equals the remaining time.
  - A spark travels down it.
- **Ticking stages**, by remaining time:

Stage	Remaining	Visual	Audio
`LIT`	4.0–2.5 s	Steady spark, slow pulse of the band lights	Tick at 2 Hz
`HOT`	2.5–1.0 s	The seams start to glow orange; faster pulse	Tick at 4 Hz, pitch rising
`CRITICAL`	1.0–0 s	White-hot seams, the body swells 8 %, it shakes, sparks spray	Tick at 10 Hz becomes a whine

- **Everyone sees every bomb.** The Carrier also gets an **overhead fuse ring**: a circular timer above the head, visible through walls within 60 studs.
- **Carrier-only feedback:**
  - a red pulsing vignette
  - a heartbeat in sync with the tick
  - the PASS prompt (§10.2)

7.3 Getting a bomb — the tag (F-BOMB-02)

- A Bomber's **touch** on a Runner who isn't carrying (hitbox overlap; see `Config.Bomb.TAG_RANGE`) sticks a **fresh 4.0 s bomb** on that Runner.
- **Bomber tag cooldown:** 1.5 s after any tag (TUNE) (`TAG_COOLDOWN`). Their hand glows again when it's ready.
- **The slap (F-BOMB-10, D-FEEL-01):** every ready Bomber holds a small copy of the bomb (the hero bomb, its face and the Bomber's skin) in their right hand. The face looks at the nearest Runner. A tag is a **slap**: a quick lunge and an overhand slap, and the hand bomb clamps onto the Runner's back. The hand is empty while the tag cools down, then a fresh bomb grows back and its fuse lights when the tag is ready. So **anyone can read at a glance whether a Bomber is loaded**.
- **The stumble:** the slap shoves the tagged Runner **6 studs** forward, away from the Bomber, over 0.25 s (`TAG_SHOVE`, `TAG_SHOVE_TIME`), with a stumble. The new Carrier lands among the other Runners instead of next to the Bomber who just tagged them.

- **A Runner can carry only one bomb.** A Bomber touching a Carrier does nothing; the touch is ignored, no new bomb.
- There is **no limit on bombs in play**: every Bomber can create one with each tag.

7.4 Passing (F-BOMB-03..05)

ID	Rule
F-BOMB-03	**Contact passes.** When a Carrier's hitbox overlaps another eligible player, the bomb jumps to them **instantly** (server-authoritative). A 0.18 s visual arc plays from back to back. There is no button: contact is the pass. This makes it mobile-first, and it means accidents are part of the comedy. **The bump (D-FEEL-01):** a pass knocks the two bodies apart, **3 studs** each, away from each other (`PASS_BUMP`), so "BUMP SOMEONE!" is literally what you see.
F-BOMB-04	**The fuse keeps its remaining time.** It never resets, never extends.
F-BOMB-05	**Eligible receivers:** anyone *not in a Safe Strip* and *not already carrying*. That includes Runners and Bombers. **No pass-back for 1.0 s:** you can't pass a bomb straight back to the player who just gave it to you (`NO_PASSBACK`). This kills ping-pong. **Receive grace of 0.2 s:** a fresh Carrier can't pass in the first 0.2 s (`RECEIVE_GRACE`). This stops same-frame bounces and gives the receiver one frame of "oh no".

7.4b The Hand-Off: passing as a 1-second duel (F-BOMB-11..14) OURS

Contact still passes (F-BOMB-03): the accidents stay part of the comedy, and nobody needs a button. On top of that, every pass is a small duel. You can see the danger, the Carrier has an attack, the receiver has a counter, and there's a way to beat the counter (D-PASS-01).

ID	Rule
F-BOMB-11	**The Danger Ring.** A ring on the ground under every Carrier shows exactly how far their pass reaches, in the bomb's stage colour. It **grows as the fuse burns: 4 studs at LIT, 4.75 at HOT, 5.5 at CRITICAL** (`PASS_RANGE_STAGE`). The server uses the same radius, so what you see is the rule. A READ THE FACE bomb's ring stays LIT-sized until it turns HOT, like its face.
F-BOMB-12	**The Lunge.** A Carrier's **Dash** is a lunge: for the dash (0.34 s), the reach stretches **+2.5 studs** toward bodies in front of them (within 60° of the dash direction, `LUNGE_REACH` , `LUNGE_CONE`), and the ring stretches into an oval that way. A pass that connects during a lunge is a **SLAM PASS**: a heavier freeze, the receiver knocked 5 studs, and its own callout. **A whiff:** if you lunged at someone (a body in that cone within the stretched reach + 3 studs when the dash started) and didn't connect, you stumble for 0.35 s at 60 % speed (`WHIFF_*`), and the dodger gets DODGE!. A dash to escape is never punished.
F-BOMB-13	**The Reject. Hard Stop just as the bomb reaches you** (within 0.25 s before contact on the server, `REJECT_WINDOW`): **REJECTED!** The bomb ricochets back to the giver, even inside the no-pass-back second, and the giver is staggered for 0.4 s (`REJECT_STAGGER`). Limits: a **4 s** cooldown per player, and **one reject per bomb per player**, so there are no endless volleys. A Carrier can't reject. **The duel:** Lunge beats running, Reject beats Lunge, and a feint beats Reject (fake the lunge, wait out their Hard Stop, then walk in while they're on cooldown).
F-BOMB-14	**The Chain Song.** Every pass in a chain plays the next note of a rising major scale, one licensed pluck pitched by semitones, so a ×5 chain sounds like a climbing arpeggio. The bomb carries its chain: a ×N tag and a hotter rim from ×2. On the transfer: a freeze on both bodies, the ricochet arc, "FROM name" on the receiver's screen, and a relief whoosh on the giver's.

Bots lunge (their pass dash) and reject with skill-tier odds, so the duel exists in a solo round. **No new Stars:** a SLAM PASS pays as a pass, and a REJECTED! pays XP only. **Success measure (pace probe):** passes per bomb from 0.64 to 1.2+, boom % within 45–60, 2–6 rejects a round, and some whiffs. First bot probe (2026-10-18): 0.83 passes per bomb, 38 % SLAM PASSES, 1.8 rejects a round, 14 whiffs in 8.5 min, boom 61 %. **Bots take a free hand-off within 7 studs** (`Config.Bots.OPPORTUNITY_RANGE`), as a player would.

7.5 Booms (F-BOMB-06..08)

ID	Case	Result
F-BOMB-06	**It blows on a Runner**	The Runner booms: they're flung by an explosion impulse, ragdoll for 1.6 s, then **become a Bomber** (respawning in the middle at the next `READY` , or getting up where they landed if the crossing is still running and the field is clear). Credit goes to the **last Bomber who tagged this bomb's chain** (`bomb.origin`).
F-BOMB-07	**It blows on a Bomber** (they received a pass)	Revenge: the Bomber is flung and **stunned for 3.0 s** (`BOMBER_STUN`); they can't tag or move. They stay a Bomber. A Bomber **can't pass a bomb on**: bombs on Bombers are locked, and they just blow.
F-BOMB-08	**Blast splash**	Everyone else within 7 studs (`SPLASH_RADIUS`) gets a knockback shove (`Characters.push` , no ragdoll, no elimination). This is comedy and chaos, and lets you blow a crowd apart. TUNE

7.6 Defusing (F-BOMB-09)

- A Carrier who enters a Safe Strip **defuses** the bomb.
  - The fuse snuffs with a hiss and a smoke curl.
  - The bomb pops off the clamp and clatters to the ground, then fades.
- **With ≤ 1.0 s left** it's a **CLUTCH DEFUSE**: slow-mo, a big callout and a big reward (§11).
- A defuse doesn't make the Runner immune to new tags on the next crossing. It's a fresh start.

7.7 Edge cases (all server rules)

Case	Rule
Two Carriers touch	Nothing happens: neither is eligible (both carrying).
A Carrier touches two players in the same frame	The pass goes to the **closest** one.
A Carrier leaves the game	The bomb booms in place at their last position: splash only, no credit.
A Bomber leaves while their bomb is in play	Credit falls back to "the field"; no Bomber gets credit.
A Carrier falls off the map	The bomb booms at the kill plane. They become a Bomber (respawn middle).
The crossing ends while bombs are live	Bombs on Runners in Safe Strips are defused. Bombs on the field boom with the field detonation.
The last Runner is carrying when they become last	The bomb keeps ticking. If it booms, the Bombers win immediately. If they defuse it, the Last Stand begins.

7.8 The bomb's numbers (Config.Bomb)

Key	Value	Meaning
`FUSE`	4.0	Seconds on a fresh bomb TUNE
`STAGE_HOT` / `STAGE_CRITICAL`	2.5 / 1.0	Stage thresholds
`TAG_RANGE`	3.5	Studs, root-to-root (Bomber reach) TUNE. Was 4.5: with the lag slack, tags landed from 6.5 studs and never looked like a touch (D-FEEL-01)
`TAG_SHOVE` / `TAG_SHOVE_TIME`	6 / 0.25	studs / s: the slap's stumble (F-BOMB-10)
`PASS_BUMP`	3	studs each: the pass knock-apart (F-BOMB-03)
`PASS_RANGE`	4.0	Studs, root-to-root (body contact) TUNE; LIT reach, see `PASS_RANGE_STAGE`
`PASS_RANGE_STAGE`	4 / 4.75 / 5.5	studs at LIT / HOT / CRITICAL: the Danger Ring (F-BOMB-11)
`LUNGE_REACH` / `LUNGE_CONE`	2.5 / 60	studs added in front during a Carrier's dash / degrees either side (F-BOMB-12)
`SLAM_KNOCK`	5	studs the receiver of a SLAM PASS is knocked
`WHIFF_TIME` / `WHIFF_SPEED`	0.35 / 0.6	s / speed while stumbling after a whiffed lunge
`REJECT_WINDOW` / `REJECT_COOLDOWN` / `REJECT_STAGGER`	0.25 / 4 / 0.4	s (F-BOMB-13)
`TAG_COOLDOWN`	1.5	s
`NO_PASSBACK`	1.0	s
`RECEIVE_GRACE`	0.2	s
`BOMBER_STUN`	3.0	s
`SPLASH_RADIUS` / `SPLASH_PUSH`	7 / 14	studs / studs of shove
`BOOM_IMPULSE_H` / `_V`	70 / 55	studs/s fling TUNE
`CLUTCH_WINDOW`	1.0	s left for CLUTCH DEFUSE
`LAG_SLACK`	2.0	Extra studs of server tolerance for contact checks (§24.1)

8. Winning, Last Stand, results

8.1 Win conditions (F-WIN-01..03)

ID	Situation	Winner	Reward key
F-WIN-01	The last Runner completes the Last Stand	**That Runner** (solo win, biggest reward in the game)	`LastStandWin`
F-WIN-02	Every Runner booms	**The Bombers.** The top Bomber by booms is MVP	`BomberWin` + `MVP`
F-WIN-03	2+ Runners survive crossing 8	**All survivors** (shared win)	`SurvivorWin`

8.2 The Last Stand (F-LAST-01..05) OURS

ID	Beat
F-LAST-01	**Lights down.** The world dims to 35 %, a spotlight locks onto the last Runner, and a drumroll starts. A banner reads "LAST STAND — " with their avatar headshot. 3 s.
F-LAST-02	**Everyone's a Bomber now.** Every Bomber lines up in the middle. A crowd chant (the audience SFX loop) builds.
F-LAST-03	**The run:** a 12 s Field Fuse. The Runner gets **+8 % speed** (`LAST_STAND_SPEED`, the "adrenaline" boost), shown as a light-streak trail. Bombers' tag cooldown stays 1.5 s.
F-LAST-04	**Win:** they reach the strip → the world snaps to full light, a fireworks volley, "VICTORY" in gold, and the camera orbits the winner. **Lose:** the boom gets a 0.5 s slow-mo, a "BOMBERS WIN" stinger, and the camera cuts to the credited Bomber.
F-LAST-05	Spectating players (every Bomber) get a **picture-in-picture** of the Runner's face cam on mobile. On PC the camera auto-follows the Runner if the player stops giving input for 2 s.

8.3 The WINNER crown (F-WIN-04)

- **One crown per server:** the round's champion wears a **gold crown with a lit sparkler** above their head for the whole next round. The champion is the Last Stand Runner; if the Bombers win, their MVP; otherwise the winner with the best round (booms ×3, clutches ×3, defuses ×2, crossings ×2, tags, passes).
- It's visible to everyone, which brings social status and earns the "Dethrone" quest (§14).

8.4 Results screen (F-RES-01..04)

ID	Element
F-RES-01	**The headline:** the winner reveal (a 3D podium with the top 3 and their avatars), or "BOMBERS WIN" with the MVP.
F-RES-02	**Your round, line by line**, counting up: crossings survived, defuses, clutches, passes, booms, win. Every line pays Stars and XP with a coin-burst.
F-RES-03	**The XP bar fills** with sound ramps. If a level-up happens it **interrupts the screen** with the full level-up celebration (§12.3).

ID	Element
F-RES-04	**The "next" hooks** (pick the 2 most relevant): "1 more round to level up", "Quest 2/3: Pass 7/10", "Gift in 3:12", "MEGA BOMB in 4:30".

8.5 A full round, as a player sees it (reference walkthrough)

16 players: 14 Runners, 2 Bombers. Crossing 1 (20 s): Bombers tag 5 Runners. Two pass their bomb to someone else, one sprints home and defuses, and two boom. *14 → 12 Runners, 4 Bombers.* **Crossing 2 (16 s):** More Bombers, more bombs in play. A Runner gets tagged, bumps a Bomber (REVENGE PASS!) and the Bomber gets stunned. Five Runners boom. → *7 Runners, 9 Bombers.* **Crossings 3–5 (13 → 10 s):** It's chaos, and the field fuse forces everyone to run. → *1 Runner left.* **LAST STAND:** Lights down, spotlight, drumroll. You run alone against 15 Bombers. You're tagged halfway with 4 s left, nobody is near enough to pass it to, so you sprint. **3... 2... 1...** You dive into the strip. **CLUTCH DEFUSE → VICTORY.** Crown on your head next round.

9. Movement & controls

9.1 Transferred as-is (F-MOVE-01)

Movement is the game's foundation, and Chase Game's system is already tuned to the user's standard. It transfers **unchanged** (§23): `MovementCore`, `MovementController`, `InputController`, `MovementValidator`, `CameraController` and `Config.Movement`.

Technique	Behaviour (Chase Game tuning, kept)
Run	Max speed 24 studs/s. The first step jumps straight to 42 % of top speed, then eases up to full over 0.95 s.
Omnimovement	The body faces the camera. W runs, A/D strafe at 85 % pace, S backpedals at 92 % with momentum.
Momentum	Letting go holds your pace for 0.14 s; a left↔right flip carries 85 % of your pace; a reversal into a backpedal plants then pushes off.
Dash	A burst covering ≈14 studs in 0.34 s, 2.2 s cooldown. Works from standstill, and backward while still facing forward.
Hard Stop + Juke	Stop dead in 0.16 s, then a 0.45 s window to snap out in any direction at 95 % speed. 0.7 s cooldown.
Slide	1.45× burst, then a glide; ≈22 % more ground than running. 2.5 s cooldown.
Slopes	Uphill −15 % at 30°, downhill +6 %.

9.2 Why the moveset matters more here (F-MOVE-02) OURS

The bomb turns every technique into a decision:

- **Hard Stop + Juke:** make a Bomber overshoot, or stop dead so a Carrier running at you overshoots its pass.
- **Dash:**
  - toward a Runner to pass the bomb before it blows (a "dash pass")
  - away from a Carrier to dodge it
  - toward the strip for a clutch defuse
- **Slide:** under a crowd, through a gap between two Bombers.

9.3 Role differences (F-MOVE-03)

	Runner	Bomber
Speed	1.00 ×	1.00 × TUNE
Dash / Slide / Hard Stop	✓	✓
Tag	—	Touch, 1.5 s cooldown
Profile key	`Profiles.Runner`	`Profiles.Bomber`
While carrying a bomb	Normal speed. No slow-down: panic should feel fast, not sluggish	Can't move while stunned

9.4 Controls (F-MOVE-04)

There is **no jump**. Space is Dash. The bomb never needs a button: contact passes it.

Action	PC	Mobile	Console
Move	WASD	Thumbstick (dynamic)	Left stick
Camera	Mouse (shift-lock on by default during rounds)	Drag on the right half	Right stick
Dash	Space	DASH button (big, bottom right)	A / Cross
Slide	C or Ctrl	SLIDE button	X / Square
Hard Stop	Shift	STOP button	B / Circle
Emote wheel (lobby)	G	Lobby button	D-pad up

Mobile button layout and sizes are in §21.3.

10. Game feel: the panic loop

10.1 The escalation ladder (F-FEEL-01)

Tension climbs on four layers at once, so players *feel* the clock without reading it:

Layer	Calm (no bomb near)	A bomb within 15 studs	You're carrying (LIT → HOT → CRITICAL)
Music	Round theme at full mix	Hi-hat layer added	The drums cut out; the heartbeat syncs with the tick
Sound	Footsteps, crowd	The nearby bomb's tick (3D, positional)	Your own tick, panned center, rising pitch; the fuse hiss
Screen	Clean	Faint warm edge glow on the bomb's side	A red vignette pulsing with the tick, growing 20 % → 45 % → 70 %
Camera	Normal	—	FOV +4° at HOT, +8° at CRITICAL; micro-shake at CRITICAL

10.2 The Carrier's toolkit (F-FEEL-02)

When you're carrying, the HUD helps you make the 1-second decision:

- **Pass targets glow:** every eligible player within 25 studs gets a soft outline (team-coloured). The nearest one gets a pulsing arrow and the word **"BUMP!"**.
- **Defuse distance:** a chevron on the screen edge points at the nearest Safe Strip. It shows a **live estimate** ("2.1 s away") in green if you can make it on the current fuse, or red if you can't.

  The estimate is your distance ÷ your current speed, a simple and honest rule of thumb. It's still a decision, not a guarantee.
- **The pass you just received** flashes the giver's name ("from Kyle") for 1 s, which seeds a revenge storyline.

10.3 Impact & punctuation (F-FEEL-03)

Moment	Treatment
Tag lands	**The slap (F-BOMB-10):** the Bomber lunges and slaps the hand bomb onto the Runner's back. A 90 ms hit-stop on both rigs, the in-house "smack" burst at the contact point (Asset Bible §5), a slap layered over the metallic "CLANK" as the clamp bites, and the bomb slams onto the back with squash-stretch, its face snapping to GOTCHA. The Runner stumbles forward. On the Bomber's own screen: a small camera punch.
Pass	A 0.18 s arc with a whistle and a motion trail. The receiver's bomb lands with the same CLANK, pitched up. The two bodies bump apart. **The Hand-Off (§7.4b):** the Chain Song note, a SLAM PASS's heavier freeze and knock, a REJECTED! ricochet.
Boom	Full explosion stack (Asset Bible §5). Camera shake scales with distance: 0.6 within 10 studs, 0.25 within 30. A ragdoll fling with a soot-stained body for 3 s.
Boom on you	0.35 s **impact freeze** (the camera holds while the explosion expands), then a whoosh as you fly. Then "YOU'RE A BOMBER" slams in with your new red role chip. You are back in control the moment you land.
Field detonation	A rolling blast wave across the field. Every on-field client gets the shake; Safe Strip players get a heavy rumble and a glass-rattle SFX.

Moment	Treatment
Defuse	The fuse snuffs with a hiss and a steam curl, the clamp releases with a CLICK, and the bomb drops and rolls. Then a soft chime.
Clutch Defuse	Everything above, plus a 0.5 s **presentation slow-mo** (the local camera, VFX and audio run at 0.3×; gameplay doesn't pause), a golden flash, and a choir hit.

Slow-mo budget (F-FEEL-04): presentation slow-mo only fires for:

- a Clutch Defuse
- a pass with ≤ 0.5 s left
- the Last Stand's end

It fires at most once every 4 s per client, so it never becomes noise.

10.4 Readability rules (F-FEEL-05) — never break these

1. **The bomb silhouette is the same for every skin.** Skins change materials, colours and trims, never the size or outline (Asset Bible §3.6).
2. **The fuse and the overhead fuse ring always show the true remaining time**, on every skin.
3. **Team colours are reserved:** Runner cyan `#2FD3FF`, Bomber red-orange `#FF4A2E`. Nothing cosmetic may use them as a dominant colour on a body.
4. **Explosion cosmetics** can change colour and shape, but keep the same timing and the same 7-stud splash ring.

10.5 The "moment" system (F-FEEL-06)

- The server keeps a 6-second **highlight buffer** of every participant's position and state, using `PositionHistory`.
- When a top-tier moment happens (Clutch Defuse, a 4+ Chain, a Last Stand finish), it's flagged.
- At `RESULTS`, the **best moment of the round** plays as a 4 s replay in the corner card ("MOMENT OF THE ROUND"), with the player's name.
- This ships in Phase P6. It reuses Chase Game's Final Moment Replay approach (§23).

11. Callouts & the reward drip

11.1 Callouts (F-CALL-01)

Every notable action pays out **instantly**, with a word, a sound and a number. Callouts stack on screen (a max of 3 at once, newest on top). Visual specs are in Asset Bible §6.

Callout	Trigger	XP	Stars	Tier
TAGGED!	You (a Bomber) stick a bomb on a Runner	5	1	Small
PASS!	You pass a bomb (capped at 10 paid per round)	5	1	Small
CLOSE ONE!	You pass with ≤ 1.0 s left	15	3	Medium
DEFUSED	You defuse with > 1.0 s left	15	3	Medium
CLUTCH DEFUSE!	You defuse with ≤ 1.0 s left	40	8	Big (slow-mo)
REVENGE PASS!	You pass a bomb onto a Bomber	20	5	Medium
REVENGE!	You pass to the exact Bomber who tagged you	+10 bonus	+3	Medium
BETRAYAL!	A Runner you passed to booms with it	10	2	Medium (comic sting)
CHAIN ×3 / ×4 / ×5+	A bomb you touched passes through 3+ Carriers	10 per link	2 per link	Big at ×5
BOOM!	A bomb you started booms on a Runner (Bomber credit)	20	3	Medium
DOUBLE BOOM / TRIPLE BOOM	2 or 3 of your bombs boom within 4 s	+25 / +60	+5 / +12	Big
SURVIVED	You end a crossing safe	10	2	Small
ALL CLEAR	Every Runner made it safe this crossing	5	1	Small (team)
DODGE!	A Bomber's tag reach passes within 1 stud of you and misses	5	0	Small
LAST STAND!	You're the last Runner	30	5	Big

Reward keys: `Config.Economy.Rewards` uses the callout name in PascalCase (e.g. `ClutchDefuse`).

Anti-farm (F-CALL-02):

- The PASS rewards cap at 10 per round.
- Passes between the same two players pay only the first time per crossing.
- Bots grant **half** XP and Stars for TAGGED, PASS and BOOM credit, so lobbies stay fair without feeling punishing.
- Exactly how this is enforced: §24.2.

11.2 The reward drip (design target, F-CALL-03)

The point of the systems in §11–§16 is that **something good is always about to happen**. Targets for an average player:

Interval	Something rewarding	Source
~10 s	A callout with a Stars or XP pop	§11.1
~2.5 min	Round results: a bar fill, a quest tick	§8.4
~5–10 min	A level-up (fast early, slower later)	§12
5 / 15 / 30 / 45 / 60 min	A playtime gift (the 30-min gift is a free crate)	§14.3
Every 15 min	A MEGA BOMB event	§16
~30 min	All 3 daily quests done → the bonus crate	§14.1
Daily	The login streak calendar	§14.2

The 30-minute session target is designed to be hit three ways at once:

- the 30-minute playtime crate
- the ~30-minute daily quest set
- two MEGA BOMB events inside any 30-minute window

They are *aligned on purpose*.

12. Progression: XP & levels

12.1 XP sources (F-PROG-01)

XP comes from the callouts (§11.1) plus the round payouts below (`Config.Economy.Rewards`):

Key	When	XP	Stars
`RoundComplete`	You finish a round (any role, any result)	15	3
`SurvivorWin`	You survive crossing 8 (a shared win)	60	15
`LastStandWin`	You win the Last Stand	150	50
`BomberWin`	You're a Bomber when the Bombers win	40	8
`MVP`	The top Bomber by booms in a Bomber win	30	10
`FirstWinOfDay`	Your first win of any kind today	100	25

Expected average: ≈ 120 XP and ≈ 30 Stars per 2.5-minute round [TUNE, verify in P4 soak].

12.2 Level curve (F-PROG-02)

`xpToNext(L) = 150 + 50 × (L - 1)`, capped at 2,000. There's no level cap; levels past 50 are shown with prestige stars.

Level	XP to next	Cumulative	≈ Time to reach (avg player)
2	150	150	1 round
5	300	900	~20 min
10	550	3,150	~65 min
20	1,100	11,650	~4 h
35	1,850	30,900	~11 h

Early levels come every ~5 minutes. That's the hook during the first session, when retention is decided.

12.3 Level-up rewards & celebration (F-PROG-03)

- **Every level:** `40 + 10 × L` Stars (capped at 300).
- **Every 5 levels:** a guaranteed cosmetic from the **Level Track**: bomb skins, fuse sparks, trails and titles that can't be bought, only earned (Asset Bible §8).
- **Celebration:**
  - A full-width banner: the level number slams in with a shockwave ring, firework pops and a choir.
  - The reward card flips over.
  - Everyone nearby sees a level-up firework on your character.
- **Level badge:** a bronze → silver → gold → platinum → obsidian frame shown on the nametag and leaderboard. It changes every 10 levels.

13. Economy: Stars

13.1 One soft currency (F-ECO-01)

Stars buy every cosmetic in the store and every crate opening. There's no second premium currency: Robux buy Stars packs or items directly (§17).

13.2 Faucets and sinks (F-ECO-02)

Faucets (per average player)	Stars/hour
Rounds and callouts	≈ 720
Daily quests (3)	≈ 230 per day
Playtime gifts	≈ 330 per day + 1–2 crates
Login streak	≈ 100–400 per day
Level-ups	≈ 150 per hour early, falling later

Sinks	Price (Stars)
Common item	400
Rare item	1,000
Epic item	2,500
Legendary item	6,000
Spark Crate	600
Inferno Crate	1,500
Quest reroll	50

13.3 The balance rule (F-ECO-03) — the lesson from Chase Game

> **No single round, event or gift may pay more than ~45 minutes of normal play's worth of Stars (≈ 550).**

In Chase Game, one Titan Hunt paid ~1,650 Stars, more than three of the cheapest Stars packs, so nobody bought Stars. Every reward table in this document respects the cap.

The P4 soak test (§26) prints the measured Stars/hour; it must land in 600–900.

14. Daily systems

14.1 Daily quests (F-DAILY-01..04)

ID	Rule
F-DAILY-01	Each day (00:00 UTC) every player gets **3 quests**: one Easy, one Medium, one Hard, drawn from the pool below without repeating yesterday's.
F-DAILY-02	**Rewards:** Easy 40 ★ + 60 XP · Medium 70 ★ + 100 XP · Hard 120 ★ + 160 XP. **All three done → a free Spark Crate opening** (the "Daily Crate").
F-DAILY-03	**One free reroll a day.** Extra rerolls cost 50 ★.
F-DAILY-04	Progress shows live in the HUD quest tracker (§21.2) and ticks with a sound and a +1 pop. A completion fires a banner and a coin shower.

Quest pool (the targets are TUNE; designed so all three take ~30 minutes):

Easy	Medium	Hard
Pass the bomb 10 times	Defuse 3 bombs	Get 2 Clutch Defuses
Survive 8 crossings	Make 3 Revenge Passes	Win 1 round (any way)
Tag 8 Runners	Get 5 Booms as a Bomber	Be part of a ×5 Chain
Play 5 rounds	Survive to crossing 4 three times	Get a Triple Boom
Dodge 5 tags	Win as a Bomber twice	Reach a Last Stand
Finish 3 crossings without being tagged	Make 2 Betrayals	Feed the MEGA BOMB 3 bombs (§16)
Play a MEGA BOMB round	Dethrone a crowned winner (boom them)	Win a Last Stand

14.2 Login streak (F-DAILY-05..07)

Day	1	2	3	4	5	6	7
Reward	100 ★	150 ★	Spark Crate	250 ★	Rare item (random, from an unowned pool)	400 ★	**This month's Streak Bomb skin** (Epic, exclusive, rotates monthly)

- **F-DAILY-06:** The calendar opens automatically on your first join each day. You claim by tapping, and the tap pays out with an animated stamp.
- **F-DAILY-07: A missed day drops you back 2 days**, not to day 1 (`Config.Daily.STREAK_MISS_PENALTY`). After day 7 it loops, and days 8+ repeat days 4–7.

14.3 Playtime gifts (F-DAILY-08..10)

ID	Rule
F-DAILY-08	Gifts unlock on **total minutes played today** (UTC day, saved), so leaving and rejoining doesn't reset progress or farm it.

ID	Rule
F-DAILY-09	**Milestones:** 5 min → 40 ★ · 15 min → 100 ★ · **30 min → a free Spark Crate** · 45 min → 150 ★ · 60 min → a Gift Box (Rare guaranteed, 15 % Epic). After that, every 30 min → 100 ★.
F-DAILY-10	The HUD gift icon shows a ring that fills as you play, and a countdown to the next gift. When a gift is ready, the icon bounces and glows. You claim by tapping, and **claiming is never automatic**, because the tap is the reward moment.

15. Streaks, leaderboards, social

15.1 Hot Streak (F-SOC-01)

- **Your streak +1** for every round you either **win** (any way) or **survive to crossing 5+**. Otherwise it resets to 0.
- **The Stars multiplier for that round's payout** (the round payouts paid at RESULTS, §12.1; callouts are not multiplied — DECISIONS D-ECO-02): streak 2 → ×1.25 · 3 → ×1.5 · 4+ → ×2.0 (`Config.Economy.STREAK_MULT`).
- **The HUD streak flame** (§21.2) grows with the streak and roars when the multiplier rises. When the streak breaks, the flame gutters out with a sad fizz. Loss aversion is the point.

15.2 Leaderboards (F-SOC-02)

Board	Scope	Stat	Reset
In-server	The player list (`leaderstats`)	Level, Wins	—
Lobby boards (3 physical boards)	Global, `OrderedDataStore`	Wins · Clutch Defuses · Booms	Weekly (Monday 00:00 UTC) + all-time tab
Top 3 weekly	Lobby	A golden statue of each top player's avatar on a plinth	Weekly

15.3 Social (F-SOC-03..05)

ID	Feature
F-SOC-03	**Squad bonus:** +10 % Stars for each Roblox friend in the same server, up to +30 %. Shown on the results screen as "SQUAD +20 %".
F-SOC-04	**Invite prompt:** after your first win of a session, a "Show your friends" button opens `SocialService:PromptGameInvite` . It's never forced or spammed (once per session).
F-SOC-05	**Group perk:** members of the game's Roblox group get the exclusive **"Insignia" fuse spark** (cosmetic) and a one-time 150 ★. Checked with `Player:IsInGroup` . **Likes and favourites are never rewarded**: incentivising votes or favourites breaks Roblox's rules.

16. MEGA BOMB event (F-MEGA-01..08) [OURS]

The scheduled event, rebuilt from Chase Game's Titan Hunt framework (schedule, giant rig scaling, cinematics, event clock, summon product). The design is new.

ID	Rule
F-MEGA-01	**Schedule:** every 15 minutes on the UTC wall clock (`:00 :15 :30 :45`). The first round to start after a slot opens is a MEGA BOMB round. The same moment applies on every server.
F-MEGA-02	**Roles:** one player becomes the **MEGA BOMBER**, a 3× giant with a huge bomb on its back and a crown fuse. Humans are weighted 3× over bots. Everyone else is a Runner. There are no regular Bombers at the start.
F-MEGA-03	**Mega tag:** the giant's touch sticks a **3.0 s** bomb (shorter than normal). Runners who boom become regular Bombers on the Mega team.
F-MEGA-04	**STOMP** (the Mega's Dash button): a ground slam, telegraphed by a ring for 0.6 s. It flings Runners within 12 studs (no bomb, no elimination; it scatters them). 6 s cooldown.
F-MEGA-05	**The Mega Fuse:** the giant bomb on its back has a 90 s fuse, which is the event clock. When it burns out, **the MEGA BOMB detonates**: a whole-map blast, and every Runner not in a Safe Strip booms. Runners in a Strip at that moment **win**.
F-MEGA-06	**Feed it bombs:** a Runner who passes their bomb onto the MEGA BOMBER **staggers it for 1.5 s and burns 6 s off the Mega Fuse** (the fuse visibly jumps). So the Runners' play is: get tagged on purpose, then feed it. It uses the existing pass mechanic, with no new controls.
F-MEGA-07	**Crossings continue as normal** during the event (the Field Fuse still runs), so Runners must keep crossing *and* time the finale.
F-MEGA-08	**Rewards:** Runners who win get ×3 round Stars (capped by F-ECO-03) and a free Spark Crate. If every Runner booms before the Mega Fuse ends, the MEGA BOMBER wins: ×3 Stars and the "Mega" title track.

Summon (F-MEGA-09): a "Summon the MEGA BOMB" developer product (§17.2) makes the next round a MEGA BOMB round with the buyer as the giant. Only one is queued per server; extra buyers queue for later rounds, and their place in the queue is shown.

The lobby and HUD event clock (F-MEGA-10) is the same widget pattern as Chase Game's Titan clock: a compact lobby card, plus a compact in-round chip in the top bar row on phones (§21.2).

17. Store, cosmetics, crates

17.1 Cosmetic categories (F-STORE-01)

Everything is cosmetic. Nothing changes the outcome of a round (§1.3). Every category is something **other players see during the most-watched moments**, because that's what sells.

Category	Shows when	Launch count	Readability rule
Bomb Skins (hero category)	Every time you carry, all round long	16	Same silhouette and fuse; materials and trims only (§10.4)
Explosions	Every boom you cause *and* your own boom	8	Same timing, same 7-stud ring
Fuse Sparks	Your bomb's spark plus your overhead fuse ring	8	The ring keeps the true time
Trails	Dash, slide and Last Stand	8	No team colours dominant
Victory Booms	Your win celebration and the podium	6	—
Titles	Nametag, leaderboard, results	20	—
Emotes	Lobby and results podium	8	Lobby only

Full item list, art briefs and rarity: Asset Bible §8.

17.2 Prices (F-STORE-02)

Rarity	Stars	Robux (direct)	Visual tier (frame, glow, reveal)
Common	400	29	Slate frame
Rare	1,000	99	Cyan frame and shimmer
Epic	2,500	249	Violet frame, particles
Legendary	6,000	449	Gold frame, animated foil, unique reveal music
Exclusive	—	—	Obsidian frame. Earned only (level track, streak, events, VIP)

Developer products:

Product	Robux	Contents
500 Stars	49	500 ★
1,200 Stars	99	1,200 ★ (+20 %), tagged **BEST VALUE**
3,000 Stars	229	3,000 ★ (+30 %)
7,000 Stars	479	7,000 ★ (+45 %)
Starter Bundle (once per account)	99	1,500 ★ + the **"Rookie Brass" Rare bomb skin** + the title "Fresh Fuse" (≈ 3× value)
Common / Rare / Epic / Legendary Item	29 / 99 / 249 / 449	The chosen item (one product per tier; the server remembers which item)

Product	Robux	Contents
Spark Crate	49	One opening
Inferno Crate	129	One opening
Summon the MEGA BOMB	99	§16

Game passes:

Pass	Robux	Contents
VIP	299	Gold nametag and chat tag, +10 % Stars, the exclusive "Gilded" bomb skin and "Champagne" explosion, and the VIP lounge in the lobby
Bomb Workshop	149	Recolour any owned bomb skin's trim and band lights from 16 curated swatches. Team colours are excluded; the swatches are tested for readability

17.3 The shop screen (F-STORE-03..05)

ID	Rule
F-STORE-03	**Featured (daily rotation):** 4 items rotate at 00:00 UTC, with one "DEAL" at 20 % off. A countdown shows the next rotation.
F-STORE-04	**Every item has a 3D preview:** a turntable in a ViewportFrame on a pedestal. Bomb skins show the full ticking cycle (LIT → HOT → CRITICAL) on a loop, and explosions play on a test dummy. **No flat icons for hero items.**
F-STORE-05	**Try before you buy:** "Preview in game" equips the item for one round in the practice range (§19) without owning it.

17.4 Crates (F-STORE-06..09)

Crate	Price	Common	Rare	Epic	Legendary
Spark Crate	600 ★ / 49 R$	60 %	30 %	9 %	1 %
Inferno Crate	1,500 ★ / 129 R$	—	55 %	35 %	10 %

ID	Rule
F-STORE-06	**The odds are always shown** on the crate card before purchase. This is a Roblox requirement for paid random items.
F-STORE-07	**Duplicates** refund 50 % of the item's Stars price, with "+500 ★ (duplicate)" on the reveal.
F-STORE-08	**Pity:** the Inferno Crate guarantees an Epic or better within any 8 openings. The counter shows on the card ("Epic+ in 3").
F-STORE-09	**Policy:** the server checks `PolicyService:GetPolicyInfoForPlayerAsync(player).ArePaidRandomItemsRestricted`. Where it's true, **buying crates (with Robux or Stars) is disabled**: the crate shows "Not available in your region". Free crate rewards still open, but give the same item from a fixed list instead of rolling.

The crate opening (F-STORE-10) is a premium reveal sequence: Asset Bible §7.4. The design has three beats:

1. The crate's fuse lights.
2. A rarity-coloured flare escapes through the lid seams (this is the tension beat; the colour hints at rarity).
3. The crate blows open with confetti-sparks, and the item rises on a beam.

17.5 The Starter Bundle offer (F-STORE-11)

- **The first session:** after the new player's **3rd round results**, show a one-time full-screen offer card with a 48 h countdown ("New Player Offer").
- **After that,** it lives in the shop with its countdown until it expires or is bought.
- **It's never shown more than once per session as a popup.**

18. Bots

18.1 Fill rules (F-BOT-01)

- **Bots fill each round to 16 participants** at `INTERMISSION`.
- **When a human joins mid-round**, a bot is removed at the next round boundary, never mid-round.
- **Bots use Chase Game's** `AvatarPool` (varied, realistic avatars) and natural names.
- **Looks (D-LOBBY-02):** the bots wear the avatars of the humans' friends, split evenly between the humans in the server (with 2 players, half the bots wear each one's friends), rebalanced at every intermission. Random dressed avatars fill in only when friends run out. Bots keep generated names, so only the look is borrowed.

18.2 Brains (F-BOT-02..05) — new, bomb-aware

Bots steer with Chase Game's `MovementCore` (the same physics as players) and run new decision logic in `BombBrain`:

ID	Role	Behaviour
F-BOT-02	**Runner**	Picks a lane using a threat map (Bombers' positions and speeds). Times its start (some go straight, some wait for a gap, but never past the Field Fuse's 50 % point). Uses dash, slide and hard stop to dodge when a Bomber commits. **Steers away from Carriers.**
F-BOT-03	**Runner carrying a bomb**	Each 0.1 s: if `timeToStrip < fuse - margin`, sprint to defuse. Otherwise pick the best pass target: the nearest reachable player, preferring a Bomber (revenge) when one is within reach. Then dash at them.
F-BOT-04	**Bomber**	Picks a target (the reachable Runner with the best intercept angle; spreads out from other Bombers); cuts off lanes; dashes to close the last 6 studs. **Avoids Carriers** (doesn't want a revenge pass).
F-BOT-05	**Skill tiers**	Rookie / Regular / Pro: reaction delay 0.35 / 0.22 / 0.12 s, decision error 25 / 12 / 5 %, and dash usage. Mix per round: 30 / 50 / 20 %. **A new player's first 2 rounds** get a 60 / 40 / 0 mix.

18.3 Fairness (F-BOT-06)

- Bots never get information players don't have: no seeing through walls, and the same reaction delays.
- Bots never get movement advantages: the server validates the same speeds.

19. Lobby & intermission

19.1 The lobby: "The Depot" (F-LOBBY-01)

A compact, premium lobby island built in the new art direction (Asset Bible §4.3). Everything is reachable in under 6 seconds of running:

Spot	Purpose
Spawn plaza	Where you land. The MEGA BOMB **clock tower** stands here: a giant fuse burns down the tower toward the next event (the world-space version of the event clock).
Shop kiosk	Opens the Shop. The featured items rotate on 3D plinths outside.
Quest board	Opens Quests. Shows today's 3 quests as physical cards pinned to the board.
Leaderboard wall	3 boards + 3 weekly champion statues.
Fuse Range (practice)	Training dummies to tag and pass to. There's a "practice bomb" dispenser and a defuse pad. New players learn here safely, and AFK players have something to do.
VIP lounge	A gold-trimmed balcony with a better view and a VIP-only firework button.

19.2 Intermission (F-LOBBY-02)

- **Length:** 12 s.
- **The HUD shows** "NEXT ROUND IN 12" and the MEGA BOMB chip.
- **When a menu is open at round start:** the menu closes gracefully, with a 0.3 s slide-out, before the teleport. No hard cuts.

20. Onboarding & the first session (F-ONB-01..07)

The first 5 minutes decide retention. Every step is measured (§25.2).

ID	Step	Detail
F-ONB-01	**Load**	A premium loading screen (Asset Bible §7.1): the game logo with a lit fuse that burns as assets load. The goal is under 6 s to playable.
F-ONB-02	**The 4 rules card**	Shown once, on first join. Four animated panels, one per rule (§7.1), each with a 2 s looping illustration. One "GOT IT" button. It can be reopened from the menu.
F-ONB-03	**First round**	A new player is **never a starting Bomber** in their first 2 rounds, and the bots lean easy (§18.2).
F-ONB-04	**Contextual tips** (once each)	First carry: "BUMP someone — or run to a safe side!" First time becoming a Bomber: "Touch Runners to stick bombs!" First Field Fuse warning: "Get across before it blows!" Big, centered, and they fade after 2.5 s.
F-ONB-05	**First results**	The XP bar fills and the player is guaranteed **level 2 by the end of round 1 or 2** (`xpToNext(1) = 150`).
F-ONB-06	**Gift and quests intro**	After round 2: the quest tracker slides in, plus "Daily quests: earn Stars!" with a tap to view.
F-ONB-07	**Starter Bundle**	After round 3 (§17.5).

21. HUD & UI

21.1 Principles (F-UI-01)

- **Mobile-first:** design at 844×390, then check at 1920×1080, 1280×720 and the 2732×2048 iPad.
- **The HUD stays out of the field of play.** Nothing in the centre third except callouts, and they fade in under 2 s.
- **Every element moves:** spring-in on appear, press-squash on tap, count-up on numbers. Specs: Asset Bible §6.
- **No emoji anywhere.** Every icon is from the custom icon set (Asset Bible §6.4).

21.2 Round HUD layout (F-UI-02)

Zone	Phone (844×390)	PC
Top-left	Roblox buttons; our menu button below them	Same
Top-centre	The **crossing chip** ("3/8") above the **HUD Field Fuse**: a horizontal burning-fuse bar, 220 px wide, with a spark at its tip	Same, 300 px
Top-right	**In Roblox's top bar row** (lesson from Chase Game): the team counter (runners / bombers icons + counts) and the MEGA BOMB chip	Under the top bar
Left edge	The quest tracker (collapsed to 3 small rings; tap to expand)	Expanded by default
Centre	Callouts (stacked, max 3); the defuse chevron at the screen edge	Same
Bottom-right	DASH (110 px), SLIDE and STOP (78 px) in an arc	Key hints on hover only
Bottom-left	The thumbstick zone (no art; Roblox's)	—
Carrier layer	Red vignette, heartbeat pulse, BUMP! arrows	Same

Lobby HUD (F-UI-03): Stars (top-right), level badge and XP bar, the Hot Streak flame, the gift ring, and the quest, shop and calendar buttons in a right-side dock. The same top-bar-row rule applies on phones.

21.3 Screens (F-UI-04)

Screen	Notes
Shop	Featured, Bomb Skins, Explosions, Sparks, Trails, Victory, Titles, Emotes, Crates and Stars tabs. 3D previews.
Locker	Equip each category, with live 3D previews on your own avatar.
Quests	3 cards, progress bars, reroll, the daily crate chest.
Daily Calendar	7 stamps, the claim animation, the streak count.
Crate Opening	Full-screen reveal (§17.4).
Results	§8.4.

Screen	Notes
Level Up	The interrupt celebration (§12.3).
Event Card	MEGA BOMB info, rewards and the summon button.
Settings	Music/SFX/UI volume, camera sensitivity, shift-lock default, reduced motion (turns off shake, slow-mo and vignette pulse), colour-blind mode (adds patterns to team chips).

21.4 Copy tone (F-UI-05)

- **Headlines:** short, loud, ALL CAPS ("CLUTCH DEFUSE!").
- **Body:** sentence case and friendly ("Bump anyone to pass it.").
- **Words:** never "kill" or "die". We say **boom**, **blown up**, **out**.
- **Every string** lives in `Config.Strings`.

22. Technical architecture

22.1 Toolchain (F-TECH-01)

Area	Setup

Sync	Rojo 7 (via Rokit), same mapping as Chase Game:
	`src/shared` → `ReplicatedStorage.Shared`
	`src/server` → `ServerScriptService.Server`
	`src/client` → `StarterPlayer.StarterPlayerScripts.Client`
	`builders` → `ServerStorage.Builders`

Format and lint	StyLua and Selene. `--!strict` in every file.
Tests	Studio MCP play-tests, plus the spec runner (`src/server/Tests`).
Rojo server	The user runs `rojo serve` ; agents never start a second one.
Assets	3D assets are authored as deterministic Blender 4.5 scripts in `art/` (Asset Bible §9). Audio sources go in `audio/` . UI art goes in `ui/` .

22.2 Server-authoritative model (F-TECH-02)

- **The server owns all gameplay truth:**
  - phases
  - roles
  - who carries which bomb, plus the bomb's `explodeAt` (server time)
  - contact checks
  - booms
  - rewards
- **Contact is detected on the server** every `Heartbeat` : an all-pairs distance check, about 120 pairs at 16 participants.
  - The client sends **no "pass" or "tag" remote**, so there's nothing to spoof.
  - The ranges include `LAG_SLACK` for replication delay (§24.1).
- **Bomb visuals are client-side.** Each client builds the bomb model locally on every Carrier from the replicated participant attributes `BombId` , `BombExplodeAt` and `BombSkin` . The fuse length is `explodeAt - workspace:GetServerTimeNow()` .
  - Nothing physical replicates.
  - Every client shows the same time.
  - The fuse burns smoothly at 60 FPS.

22.3 Modules (F-TECH-04)

Server services (`src/server/Services`):

Service	Status	Job
Net, GameState, Participants, PlayerService, DataService, AdminService, PositionHistory	**Transfer**	Plumbing (§23)

Service	Status	Job
Characters	**Transfer, trimmed**	Spawn, ragdoll, push, settle, scale
MovementValidator	**Transfer**	Speed and teleport checks
MoveService	**New, extracted**	The Dash / Dive / Look / MovementTech remote handlers, lifted out of Chase's Combat.luau
ArenaService	**Transfer, adapted**	The map contract: Safe Strips, Field, lines (from Chase's `Arena`)
RoundService	New	The state machine (§5) and win conditions (§8)
BombService	New	Tag, pass, fuse, boom, defuse and every edge case (§7). Pure rules live in `Logic/BombRules.luau`
FieldFuseService	New	The crossing timer and field detonation (§6.2)
RewardService	New	Callouts → XP/Stars, anti-farm, multipliers (§11, §15.1)
ProgressionService	Adapted	Levels and level rewards (§12)
QuestService, DailyService	New	Quests, login streak, playtime gifts (§14)
LeaderboardService	Adapted	Global and weekly boards, statues (§15.2)
StoreService, CosmeticService, CrateService, ProductService	Adapted (Chase patterns)	Shop, inventory, crates with pity and policy, receipts (§17)
MegaBombService	New (built from TitanService's schedule, scaling and cinematics)	§16
BotService + `Logic/BombBrain`	New brain; movement transferred	§18
AnalyticsService	Adapted	§25

Client controllers (`src/client/Controllers`):

Controller	Status	Job
Movement, Input, Camera, Pose, Footstep	**Transfer**	Feel (§9)
BombController	New	The bomb model on backs, fuse burn, stages, pass arcs, the overhead ring, carrier toolkit (§7.2, §10.2)
FuseController	New	The world fuse cords and HUD Field Fuse (§6.2)
FxController	New	Explosions, defuses, field detonation, slow-mo budget (§10.3)
CalloutController	New	§11.1
AudioController	New	Layered music and escalation (§10.1)
HudController, LobbyHudController	New	§21
Results, Shop, Locker, Quests, Daily, Crate, MegaBomb, Onboarding, Settings	New	§8.4, §17, §14, §16, §20, §21

22.4 Remotes (F-TECH-03)

Declared in `Shared/Net.luau`:

Remote	Direction	Payload
`RequestDash`, `RequestSlide`, `RequestHardStop`, `Look`, `MovementTech`	C→S	Transferred movement intents
`BombFx`	S→C	`{kind = "Stick" | "Pass" | "Boom" | "Defuse", bomb, from, to, pos, at, clutch?}`
`FieldFx`	S→C	`{kind = "Warn" | "Detonate" | "Clear", crossing, at}`
`Callout`	S→C (one player)	`{key, xp, stars, mult}`
`RoundFx`	S→C	Intro, Last Stand and results beats; `MoleReveal` (§6.4)
`ModFx`	S→C (one player)	`{kind = "Mole"}`: you're the Mole (§6.4, nobody else is told)
`Strike`	C→S	The Mole's STRIKE intent (§6.4); validated on the server
`ClaimGift`, `ClaimLogin`, `RerollQuest`, `Equip`, `BuyStars`, `BuyRobux`, `OpenCrate`, `MegaSummon`	C→S	Every one is validated and rate-limited

23. Transfer plan (from Chase Game)

The user's rule: bring over **movement, animations and the map (temporary)**; everything visible is new. Plumbing and bot movement are transferred too (decisions D7, D8).

23.1 Transfer list

System	Files (Chase Game paths)	Notes
Movement	`src/shared/MovementCore.luau`, `src/shared/AbilityMotion.luau`, `src/shared/Config/Movement.luau`, `src/client/Controllers/{MovementController, InputController, CameraController}.luau`, `src/server/Services/MovementValidator.luau`, `src/server/Tests/MovementCore.spec.luau`	Strip `DashCourseLayout` and the Spring ability references. Rename `Profiles.Catcher` to `Bomber`.
Movement remote handlers	From `src/server/Services/Combat.luau` (≈ lines 502–600)	Extract into the new `MoveService`. Chase's Combat file itself does **not** transfer.
Animations	`src/shared/Locomotion.luau` (with the 2026-10-15 self-heal), `src/client/Controllers/{PoseController, FootstepController}.luau`, the `Assets.Anim` movement ids, `tools/anim/` (the whole baker pipeline)	Drop the Titan variant until MEGA BOMB (P6). **The "ANIM PACK" in Chase's place is a competitor rip: never copy, port or upload it.**
Hit-stop	The fixed `hitStop` registry from Chase's `FeedbackController`	Only that function. It's the overlap-safe version from 2026-10-15.
Ragdoll	The `ragdoll`, `push`, `settle`, `unragdoll` and `setScale` parts of `src/server/Services/Characters.luau`; `CosmeticLooks` stubbed	
Map (temporary)	`builders/{MapKit, MapMeadow, Util, Environment}.luau`, Theme (only the map colours), `src/server/Services/Arena.luau`, `Config/{Arena, Maps}.luau`	A placeholder until the launch map (Asset Bible §4)
Plumbing	`src/shared/{Net, Types}.luau`, `src/shared/Util/{Signal, Trove, Rng, Log, Format}.luau`, `src/client/ClientState.luau`, `src/server/Services/{GameState, Participants, PlayerService, PositionHistory, DataService, AdminService}.luau`, `src/server/Logic/{SessionLock, ProfileSchema}.luau`, `src/server/Tests/Runner.luau`, both `init` scripts	Trim the `ORDER` lists and `Config/init` to the transferred modules. The profile schema starts fresh (§24.3).
Bots (movement + avatars)	`src/server/Services/AvatarPool.luau`, the steering parts of `BotService.luau`	The brains are rewritten (§18)
MEGA BOMB base (P6)	`TitanService` schedule, giant scaling and summon queue; `TitanController` clock widgets	Rebuilt with the new design and new art

23.2 Not transferred

Combat rules, Titan Hunt (as designed), abilities, modifiers, hazards, blocks, Brave or Bail, the cosmetics catalogue, the store UI, the crates' content, Theme/Style UI tokens, every UI controller, every texture and model, the Art3D library, the studded look.

23.3 Transfer procedure (Phase P0)

1. Create the new place (Studio → New → Baseplate) and save it as "FUSE VALLEY".
2. Copy the files in §23.1 into the new repo with the **same relative paths**.
3. Trim the ORDER lists, `Config/init`, `Net` remote names and `Assets` to what exists. Run `selene` and fix every unresolved require.
4. Run the map builders in the new place (builders/README snippet), then save.
5. **Studio check:** join, run, dash, slide, hard stop and strafe on the Meadow field. Animations play, and there are 0 console errors.

23.4 Ownership (decision D9)

- **The game lives under the same Roblox group as Chase Game.** The published animation ids and map prop assets are group-owned and work there without re-uploading.
- **If the owner ever changes,** re-publish the clips from `tools/anim/out` and update `Assets.Anim`.

24. Security, performance, data

24.1 Security (F-SEC-01..05)

ID	Rule
F-SEC-01	**No client remote can cause a tag, pass, boom or reward.** These come only from server contact checks and server timers.
F-SEC-02	**The contact range** is `TAG_RANGE` / `PASS_RANGE` + `LAG_SLACK` (2 studs), measured on server-side positions. The movement validator rejects teleports and speed hacks, so the slack can't be abused for long-range passes.
F-SEC-03	**Every C→S remote** goes through `Net`'s type check and token-bucket rate limits (transferred).
F-SEC-04	**Receipts are idempotent** (`ProcessReceipt` + a stored `purchaseId` set, Chase Game's pattern). Purchases grant only after the profile save succeeds.
F-SEC-05	**Anti-farm** (§11.1 F-CALL-02): the pair limit is `(giver, receiver, crossing)`; the PASS pay cap is 10 per round; a bot-credit halving applies. Private servers with fewer than 4 humans: Stars from callouts are halved.

24.2 Performance budgets (F-PERF-01)

Budget	Limit
Target device	60 FPS on a mid-range phone (iPhone 11 / Galaxy A52 class)
Bomb model	≤ 2,500 tris, ≤ 4 MeshParts, 1 SurfaceAppearance set (shared by skins via texture swaps), 1 PointLight (CRITICAL only)
A full explosion	≤ 220 particles at peak, ≤ 1.2 s of emission, 1 light flash
Simultaneous full explosions	3. Beyond that, the "light" variant (40 % particles) plays
Screen-space UI	≤ 2 nested ViewportFrames visible at once (shop previews paginate)
Map	≤ 8,000 parts, `StreamingEnabled` off (small map, instant visibility matters for a tag game)
Server	≤ 4 ms per Heartbeat at 30 participants (contact checks are ~270 distance checks)

24.3 Data (F-DATA-01)

The session-locked profile (transferred DataService) uses a fresh schema `v1`:

```
profile = {
  version = 1, createdAt,
  level, xp, stars,
  inventory = { [itemId] = true },
  equipped = { BombSkin, Explosion, FuseSpark, Trail, VictoryBoom, Title, Emotes = {…} },
  stats = { rounds, wins, lastStandWins, defuses, clutches, passes, booms, revenge, chains5 },
  daily = { day, quests = {{id, progress, done}}, rerolls, playSec, giftsClaimed = {} },
  login = { lastDay, streakDay },
  hotStreak, pity = { Inferno = 0 },
  flags = { tutorial, starterOffered, starterBought, groupPerk },
  receipts = { [purchaseId] = true },   -- trimmed to the last 200
}
```

25. Analytics, KPIs, QA

25.1 KPI targets (F-KPI-01)

These are our goals (not industry facts). They're used to decide **when to buy ads**:

KPI	Soft launch target	Before buying ads
D1 retention	≥ 15 %	**≥ 20 %**
D7 retention	≥ 4 %	≥ 6 %
Average session length	≥ 15 min	**≥ 25 min** (goal 30+)
Rounds per session	≥ 6	≥ 10
Payer conversion (of DAU)	≥ 0.5 %	≥ 1.5 %

The ad rule: no paid ads until the "before buying ads" column is met in organic play. This is the lesson from Chase Game's $50 test.

25.2 Events (F-KPI-02)

Built on Roblox `AnalyticsService`:

- **The onboarding funnel** (`LogOnboardingFunnelStepEvent`), in order:
  1. Loaded
  2. Rules card closed
  3. Round 1 started
  4. First carry
  5. First pass or defuse
  6. Round 1 finished
  7. Level 2
  8. Round 3 finished
  9. Quest tracker opened
  10. Round 5 finished
- **Economy** (`LogEconomyEvent`): every Stars source and sink, with the transaction type.
- **Custom** (`LogCustomEvent`): round length, crossings, booms per round, clutches, Last Stand reached/won, starter offer shown/bought, MEGA BOMB participation, playtime gift claimed (per milestone).

25.3 QA (F-QA-01)

Specs (pure modules), run with `RUN_TESTS`:

- `BombRules`: eligibility, no-passback, receive grace, closest-target tie-break, every §7.7 edge case.
- `FieldFuse` timings.
- `Rewards` math with multipliers and caps.
- `LevelCurve`.
- `Quests` progress.
- `HotStreak`.
- `CrateRoll`: odds over 100k rolls within ±0.5 %, pity, policy branch.

- `ProfileSchema` migration.

Other checks:

- **The soak test** (from P2): `SIM_MODE`, 15 bots, `TIME_SCALE 2`, 30 rounds. Pass = 0 errors and < 10 % memory growth. From P4 it also prints Stars/hour, which must land in 600–900.
- **Device pass** (from P3): a phone at 844×390 (Studio emulator + `CG_ForceDevice`) and a real-device test before each phase gate.

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26. Build phases & Definition of Done

26.1 Phases

Phase	Goal	Contents	Definition of Done
P0 Setup	The foundations run	§23 transfer, repo, place, Rojo, temporary map	Full movement and animations on the Meadow field; 0 console errors; `selene` clean
P1 Core loop	A playable round	RoundService, ArenaService, BombService (all of §7), FieldFuse, Last Stand, win conditions, a grey-box HUD, ragdoll booms	A full round plays end to end with 2+ players in Studio Team Test; every §7.7 edge case passes its spec
P2 Bots	A solo player gets a full round	BombBrain (Runner, Carrier, Bomber), fill rules, skill tiers	A solo round is lively; bots defuse and pass sensibly; the soak test passes
P3 Feel & hero assets	It feels incredible	**The hero bomb** (Asset Bible §3), the explosion stack (§5), the audio escalation (§10.1), callouts (§11), Last Stand presentation, results screen, the launch map's first pass	Every shipped asset passes the Quality Bar checklist (Asset Bible §1.3); 60 FPS on a mid-range phone; the user signs off on the bomb
P4 Retention	The 30-minute loop	XP and levels, Stars, quests, login streak, playtime gifts, Hot Streak, leaderboards, squad bonus, onboarding	Economy soak lands at 600–900 Stars/hour; every funnel event fires; a new player hits level 2 within 2 rounds
P5 Store	People can pay	Shop, Locker, crates (pity, policy), products, passes, Starter Bundle, the launch catalogue (all premium)	Studio test purchases work end to end; receipts are idempotent; odds are shown; the policy branch is tested
P6 Event & launch	Ship	MEGA BOMB, round modifiers (§6.4), final lobby, final map, moment replay, settings and accessibility, analytics dashboards, performance pass, store page (icon, thumbnails, trailer frames)	The launch checklist passes, and soft launch begins (KPIs in §25.1)

26.2 Working rules (same discipline as Chase Game)

- Every task ends with:
  - `stylua src`
  - `selene src`
  - `rojo build`
  - a Studio play-test with 0 errors
  - `docs/PROGRESS.md` ticked
  - a commit `P<n> <TaskID>: <summary>`
- **Pull before work**; the partner works in the same repo.
- **Play-tests that touch the real profile:** record the profile with `data` first and restore it afterwards (the Chase Game lesson).

Appendix A. Config index

Module	Key values (defaults in this document)
`Config.Game`	`NAME = "FUSE VALLEY"`, `MAX_PLAYERS = 30`
`Config.Round`	`INTERMISSION = 12`, `INTRO = 4`, `READY = 2`, `RESOLVE = 1.5`, `RESULTS = 8`, `MAX_CROSSINGS = 8`, `FIELD_FUSE = {20,16,13,11,10,10,10,10}`, `LAST_STAND_FUSE = 12`, `LAST_STAND_SPEED = 1.08`, `LAST_STAND_MIN_RUNNERS = 4`, `BOMBERS_PER = 8`, `BOMBERS_MAX = 3`, `BOMBER_REPEAT_GUARD = true`, `IDLE_WARN = 5`, `FUSE_WARN = 5`, `FUSE_PULSE = 3`
`Config.Bomb`	§7.8
`Config.Movement`	Transferred unchanged (§9.1); `Profiles.Bomber`
`Config.Economy`	`Rewards` (§11.1, §12.1), `PASS_PAY_CAP = 10`, `BOT_CREDIT_MULT = 0.5`, `STREAK_MULT = {[2]=1.25,[3]=1.5,[4]=2.0}`, `SQUAD_BONUS = 0.1`, `SQUAD_MAX = 0.3`, `ROUND_STAR_CAP = 550`, `RARITY_PRICES = {Common=400, Rare=1000, Epic=2500, Legendary=6000}`, `VIP_STARS_MULT = 1.10`
`Config.Progression`	`XP_BASE = 150`, `XP_STEP = 50`, `XP_CAP = 2000`, `LEVEL_STARS = {base=40, per=10, cap=300}`, `TRACK_EVERY = 5`
`Config.Daily`	`QUESTS` (pool, §14.1), `QUEST_REWARDS`, `REROLL_COST = 50`, `LOGIN_REWARDS` (§14.2), `STREAK_MISS_PENALTY = 2`, `GIFTS` (§14.3)
`Config.Mega`	`SCALE = 3`, `HUMAN_WEIGHT = 3`, `TAG_FUSE = 3.0`, `STOMP_RADIUS = 12`, `STOMP_TELEGRAPH = 0.6`, `STOMP_CD = 6`, `MEGA_FUSE = 90`, `FEED_BURN = 6`, `FEED_STAGGER = 1.5`, `REWARD_MULT = 3`
`Config.Store`	`Products` (§17.2), `Passes`, `Crates` (odds, pity = 8), `FEATURED = 4`, `DEAL_DISCOUNT = 0.2`, `STARTER_OFFER_ROUND = 3`, `STARTER_OFFER_HOURS = 48`
`Config.Modifiers`	`EVERY = 3`, `POOL = {Face, LightsOut, Mole}`, Face: `FUSE_MIN = 3.0`, `FUSE_MAX = 5.0`; Lights Out: `LAMP = {16, 20, 24}`, `RUNNER_GLOW = 6`, `EYE_RANGE = 28`; Mole: `MIN_RUNNERS = 4`, `MIN_ROUNDS = 5`, `STRIKE_TIME = 3`, `STRIKE_RETRY = 5`, `DEADLINE = 3` (§6.4)
`Config.Bots`	`FILL_TARGET = 16`, `TIERS` (§18.2), `TIER_MIX = {0.3,0.5,0.2}`, `NEWBIE_MIX = {0.6,0.4,0}`
`Config.Strings`	Every player-facing string